

MAT7058 Combinatorics II, Exercises 4

Due Date: 13 April 2026, Monday, beginning of class.
Only handwritten paper submissions in English are accepted.

- Each exercise is worth 4 points.
- In general, you have to prove everything and only use things which were introduced in class. Ask if you are unsure.

- 1 Construct a Hadamard matrix of order 28 using Williamson's method.
- 2 Suppose that M is an $m \times n$ matrix with 0, 1-entries such that the Hamming distance between any two distinct rows is at least d , that is, any two rows differ in at least d positions.
 - (a) Count triples $(i, j, k) \in [m]^2 \times [n]$ with $i \neq j$ and $M(i, k) \neq M(j, k)$ in two ways. Conclude that if $2d > n$, then

$$m \leq \frac{2d}{2d - n}.$$

This is known as *Plotkin's Bound* in coding theory.

- (b) Characterize equality in Plotkin's Bound.
 - (c) Suppose that $d = n/2$. Prove that $m \leq 2n$ and show that equality holds if and only if there exists a Hadamard matrix of order n .
- 3 For an Abelian group G of order v , we call a k -subset $\mathcal{D} \subseteq G$ a (v, k, λ) -*difference set* if each nonzero element $g \in G$ occurs exactly λ times as a difference $x - y$ for all $x, y \in \mathcal{D}$.
 - (a) Let $q = 4n - 1$ be a prime power. Show that the set of nonzero squares in $\mathbb{F}_q$ is a $(4n - 1, 2n - 1, n - 1)$ -difference set in the additive group $\mathbb{F}_q$.
 - (b) A difference set with parameters $(4n - 1, 2n - 1, n - 1)$ is called *Hadamard difference set*. Show that any Hadamard difference set gives rise to a Hadamard matrix.
 - 4 An *affine plane* is a rank 2 incidence geometry $\mathcal{A} = (\mathcal{P}, \mathcal{L}, I)$ that satisfies the following axioms:
 - (AP1) Through any two distinct points there is exactly one line.

(AP2) If L is a line and P is a point outside of G , then there is precisely one line through P that has no point common with L . (Playfair's parallel axiom.)

(AP3) There exist three noncollinear points.

(a) Let L_∞ be a line of a projective plane $\mathcal{G} = (\mathcal{P}, \mathcal{L}, I)$. Denote the set of points incident with L_∞ by $\mathcal{P}_\infty$. Show that $\mathcal{A} = (\mathcal{P} \setminus \mathcal{P}_\infty, \mathcal{L} \setminus \{L_\infty\}, I^*)$ is an affine plane. Here I^* denotes the natural restriction of I to $\mathcal{A}$.

(b) In an affine plane, we call two lines K, L *parallel* if they do not intersect.¹ Show that “being parallel” is an equivalence relation. We call each equivalence class *parallel class*.

(c) Given an affine plane $\mathcal{A} = (\mathcal{P}^*, \mathcal{L}^*, I^*)$, let $\mathcal{P}$ be $\mathcal{P}^*$ and the parallel classes, and let $\mathcal{L}$ be $\mathcal{L}^*$ and one further line L_∞ . Furthermore, define I as follows:

- i. $P I L$ if and only if $P I^* L$ for $P \in \mathcal{P}^*, L \in \mathcal{L}^*$,
- ii. $P I L_\infty$ for no point $P \in \mathcal{P}^*$,
- iii. $\Pi I L$ if and only if $L \in \Pi$ for any parallel class Π and all $L \in \mathcal{L}^*$,
- iv. $\Pi I L_\infty$ for any parallel class Π .

Show that $\mathcal{G} = (\mathcal{P}, \mathcal{L}, I)$ is a projective plane.

5 Let $\mathcal{A} = (\mathcal{P}, \mathcal{L}, I)$ be a rank 2 incidence geometry with $\mathcal{P} = \mathbb{R}^2$, $\mathcal{L} = (\mathbb{R} \cup \{\infty\}) \times \mathbb{R}$, and $P = (x, y)$ incident with $L = (m, b)$ for

- (a) $x = b$ if $m = \infty$,
- (b) $y = \frac{1}{2}mx + b$ if $m \leq 0, x \leq 0$,
- (c) $y = mx + b$ if $m \geq 0$ or $x \geq 0$.

Show that $\mathcal{A}$ is an affine plane. The plane $\mathcal{A}$ is called the *Moulton plane*.

6 Let $\mathcal{O}$ be an oval in the projective plane of order n , n even. Show that every point not in $\mathcal{O}$ is incident with precisely 1 or $q + 1$ tangents.

¹This definition is no good in higher dimension. (Why?) We use it here to keep life simple.